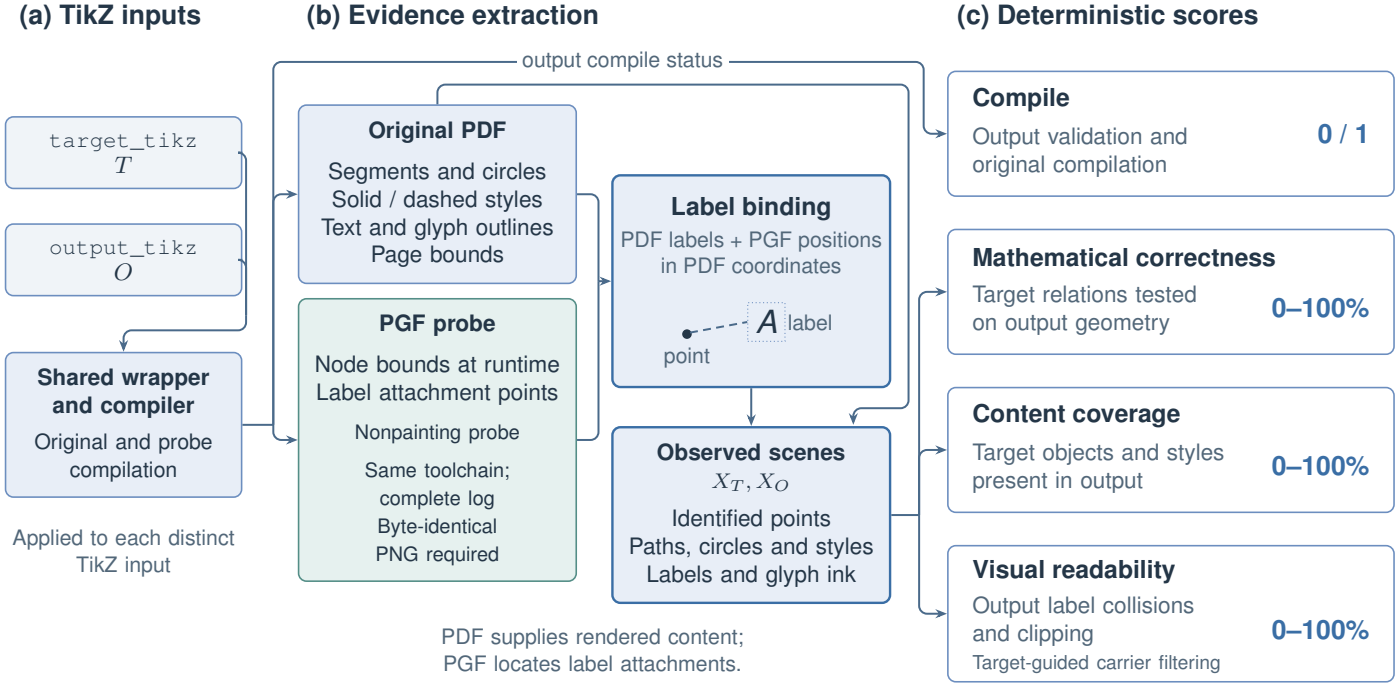
(a) TikZ inputs
(b) Evidence extraction
(c) Deterministic scores
output compile status
target_tikz
T
output_tikz
O
Shared wrapper and compiler
Original and probe compilation
Applied to each distinct TikZ input
Original PDF
Segments and circles
Solid / dashed styles
Text and glyph outlines
Page bounds
PGF probe
Node bounds at runtime
Label attachment points
Nonpainting probe
Same toolchain; complete log
Byte-identical PNG required
Label binding
PDF labels + PGF positions in PDF coordinates
A label
point
Observed scenes
X_T, X_O
Identified points
Paths, circles and styles
Labels and glyph ink
PDF supplies rendered content;
PGF locates label attachments.
Compile
Output validation and original compilation
0 / 1
Mathematical correctness
Target relations tested on output geometry
0–100%
Content coverage
Target objects and styles present in output
0–100%
Visual readability
Output label collisions and clipping
Target-guided carrier filtering
0–100%
A valid target is required. Failed output compilation gives four zero scores.